

LP Graphical Worksheet — Problem Q2

Minimize $C = 3x + 5y$

Subject to:

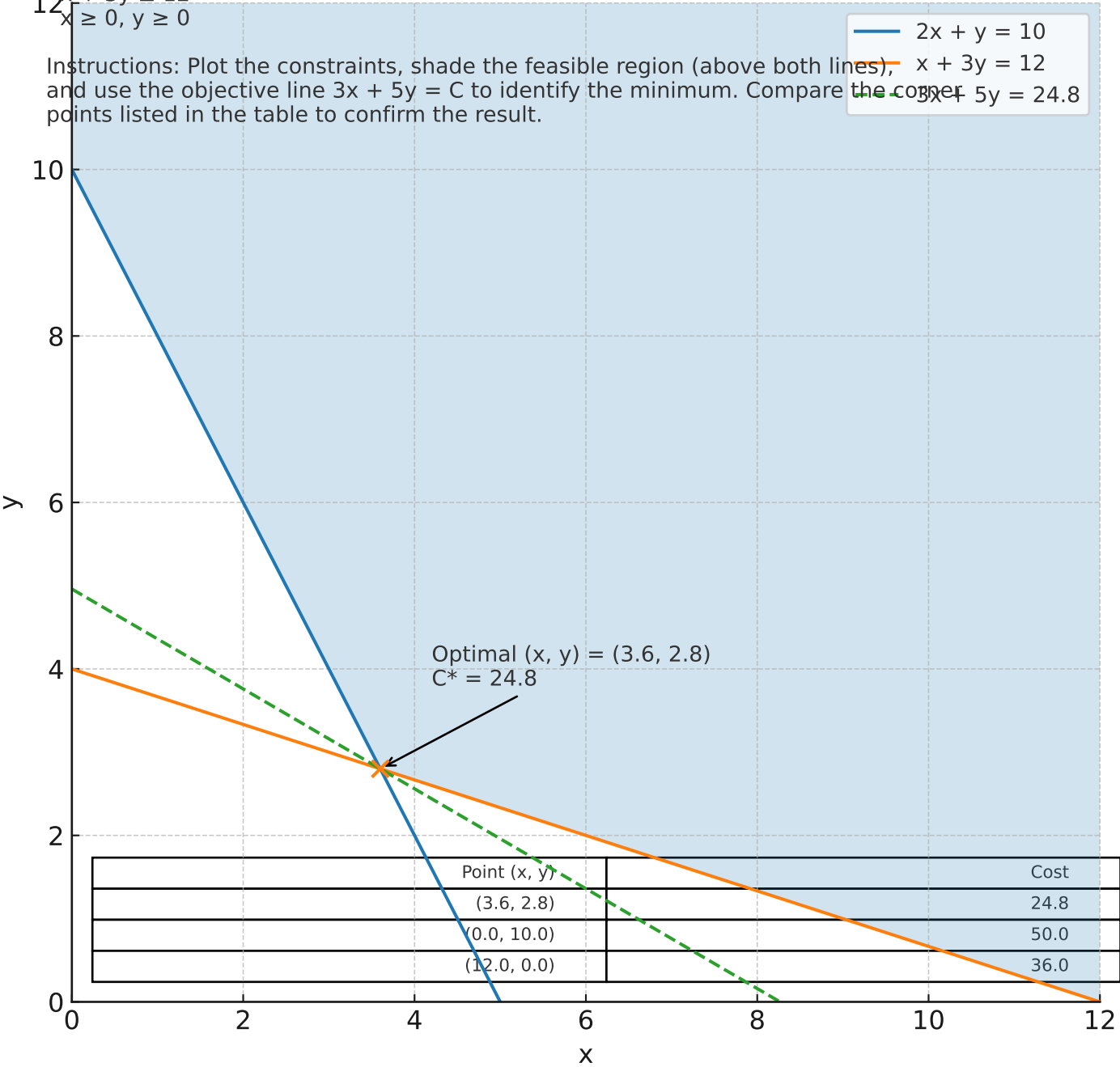
$2x + y \geq 10$

$x + 3y \geq 12$

$x \geq 0, y \geq 0$

Feasible Region & Optimum

Instructions: Plot the constraints, shade the feasible region (above both lines), and use the objective line $3x + 5y = C$ to identify the minimum. Compare the corner points listed in the table to confirm the result.



Optimal solution: $(x, y) = (3.6, 2.8), C_{\min} = 24.8$